

**Mohammed Reabai J 250701435-2025-CSE** ▾**M2****Started on** Friday, 25 September 2026, 3:28 PM**State** Finished**Completed on** Friday, 25 September 2026, 3:29 PM**Time taken** 22 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that $A[j] - A[i] = k$, $i \neq j$.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as $5 - 1 = 4$

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2
3  int main() {
4      int n, k;
5      scanf("%d", &n);
6
7      int a[n];
8
9      for (int i = 0; i < n; i++) {
10         scanf("%d", &a[i]);
11     }
12
13     scanf("%d", &k);
14
15     int i = 0, j = 1;
16
17     while (i < n && j < n) {
18         if (i == j) {
19             j++;
20         }
21         else if (a[j] - a[i] == k) {
22             printf("1");
23             return 0;
24         }
25         else if (a[j] - a[i] < k) {
26             j++;
27         }
28         else {
29             i++;
30         }
31     }
32
33     printf("0");
34
35     return 0;
36 }
37

```

	Input	Expected	Got	
✓	3 1 3 5 4	1	1	✓
✓	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✓
✓	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✓
✓	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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